

CHAPTER 9

ROLE OF SPIKE ENERGY, HFD AND SHOCK PULSE (SPM) & SPECIFICATION OF THEIR ALARM LEVELS AT VARIOUS SPEEDS

9.1 SPIKE ENERGY AND SHOCK PULSE:

Ultrasonic instruments are designed to measure energy levels in roughly the 25,000 Hz to 100,000 Hz frequency range (1,500,000 - 6,000,000 CPM). These include *Spike Energy*¹ and *Shock Pulse*². Each of these two parameters were developed in the 1970's and are designed to mechanically and electrically respond only to high frequency problem sources which excite the resonant frequency of their mounted transducer. Note the effective range for Spike Energy is from 5,000 Hz to 60,000 Hz as shown in Illustration A. In the case of spike energy, one of the most common accelerometers now in use (*IRD 970*) has a mounted natural frequency of about 30,000 Hz. Similarly, the reference mass within the shock pulse transducer is designed to respond at its resonant frequency of approximately 32,000 Hz.

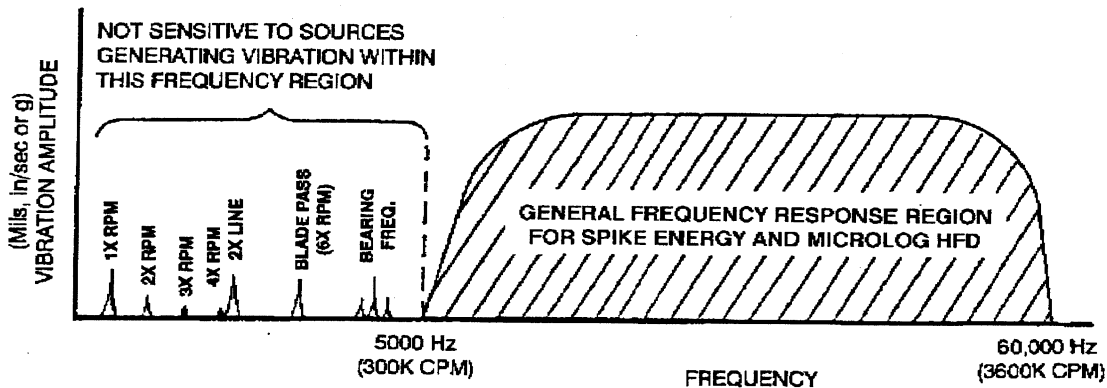


ILLUSTRATION A.
GENERAL FREQUENCY RESPONSE CURVE
FOR SPIKE ENERGY AND MICROLOG HFD

Each of these technologies has proven worthwhile and capable of acting as effective tools at picking up initial stages of bearing wear if the user follows very strict measurement rules (these are covered in a Vibration Institute article entitled "The Use of Spike Energy for Fault Analysis and Machine Condition Monitoring" written by Joseph M. Shea and James K. Taylor of IRD). When bearings first begin to wear, they begin to excite natural frequencies of their components themselves (rolling elements, bearing races and cage) as discussed in the special rolling element bearing section in Chapter 6, Section 6.09. One set of bearing component natural frequencies is concentrated within the range of 500 to 2000 Hz (30,000 to 120,000 CPM). Another set is found within ultrasonic frequency ranges near the SPM and spike energy transducer natural frequencies. Therefore, when incipient wear just begins within rolling element bearings, the bearing components begin to impact, exciting these natural frequencies which likewise excite the SPM and spike energy transducer natural frequencies. Basic defect mechanisms which generate ultrasonic response include:

- a. Micro spalls and cracks from fatigue or overstressing caused by brinelling, false brinelling, misalignment, overload, incorrect sealing and improper fits.
- b. Surface roughening from lack of sufficient lubrication.
- c. Surface indentations from hard contaminant micropitting from electric current passing through the bearing.

In the case of spike energy, accelerometers have been designed which have mounted natural frequencies on the order of 30,000 Hz (1,800,000 CPM) in the ultrasonic range. The sharp pulses and broadband random ultrasonic excitation from the bearings will excite the accelerometer natural frequency whether the accelerometer is stud-mounted, magnet-mounted, or probe-mounted (obviously, the stud-mount gives the highest spike energy magnitudes and most repeatable results). But all three mountings will provide spike energy measurements. This is somewhat surprising when one recalls what happens to vibration measurement quality results when he goes from stud to magnet to probe mount. However, like the empirical data that has been captured on rolling element bearings showing that their set of ultrasonic natural frequencies shows little change whether "free-free" or with the bearings mounted in the machine, the same appears to be true in the case of spike energy. In "layman's" terms, spike energy is generated when the incoming ultrasonic frequencies excite the accelerometer natural frequency, an electrical voltage response occurs which is then electronically conditioned and filtered, converting the end product to what might be called "impact energy" which is proportional to the incoming impact velocity. It is probably best thought of as a friction measuring parameter. Figure 1 shows a spike energy time waveform which would be seen by measurement on a faulty rolling element bearing.

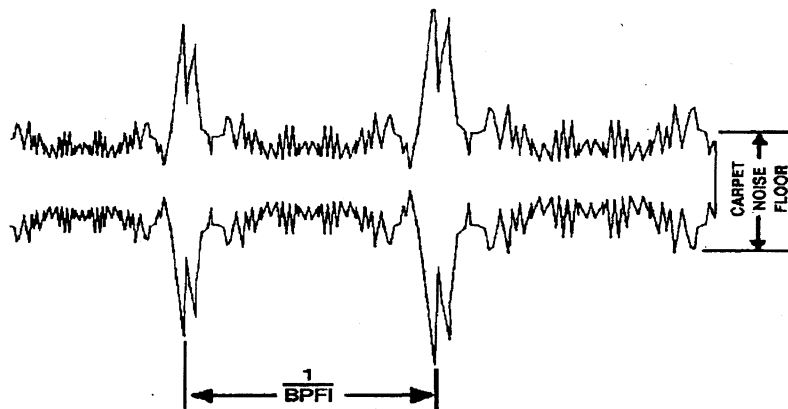


FIGURE 1
SPIKE ENERGY TIME WAVEFORM FOR A
FAULTY ROLLING ELEMENT BEARING

These pulses acting at high frequencies propagate rapidly through the structure. In the case of bearings, these elastic wave energy pulses travel through the bearing and surrounding structure at velocities approaching 16,000 ft/sec which is roughly the velocity of sound elastic waves in steel (10,900 mi/hr). When these impact waves encounter an interface of two materials or components, they lose much of their energy. This is shown in Figure 2. This interface may be a region between the bearing outer race and its housing or between one machine part and another to which this part is fastened. When these ultrasonic waves encounter an interface, some of the energy is transmitted through, and some is reflected back towards the source. The amount of energy reflected depends on a number of factors: the sharpness of the pulse, the difference in metals at the interface, how tightly the sources mate together, etc. In general, it is estimated that typically approximately 60% to 80% of the energy is reflected at each interface. **THEREFORE, THIS EXPLAINS WHY IT IS SO IMPORTANT TO KEEP THE MEASURING ULTRASONIC TRANSDUCER AS CLOSE AS POSSIBLE TO THE BEARING BEING EVALUATED.**

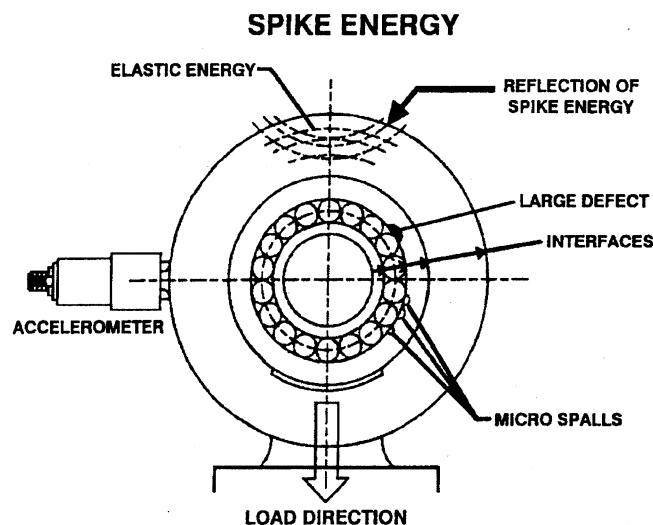


FIGURE 2
REFLECTION OF SPIKE ENERGY AT COMPONENT INTERFACES

While this energy loss may be considered as a weakness, it also has an advantage. For example, when a bearing begins to fail, much of its vibration not only will exist within this bearing, but will transmit to other structures and bearings. On the other hand, ultrasonic energy is much more localized, making it easier to isolate the specific bearing having the problem.

Here, it is important to point out that not only does spike energy and shock pulse respond to bearing wear, but also such ultrasonic measurements respond to each of the following:

- bearing wear (as stated before)
- bearing lubrication
- cavitation
- rotor or seal rub
- belt squeal
- gear meshing

- sheave rub against a guard
- impact excitation of machine parts (i.e., valve in a reciprocating machine)
- steam and high pressure air flow

Since they do respond to all of these problems, this makes ultrasonic measurements of even greater use. In some cases, it might be confusing whether the problem is radiating from a bearing or from one of the other many sources listed above. However, if one remembers that ultrasonic energy dissipates rapidly (is very localized), it can help him in diagnosing which of several problems might exist. For example, in the case of a pump, if spike energy is high on both pump bearings, it is very likely that the pump might be experiencing cavitation or a seal rub. If the operator likewise records that he hears something like a "gravel sound", it might make a strong case for cavitation. On the other hand, if on the same pump, spike energy were high only on one of the two bearings, it might suggest possible wear or lubrication problems with this one bearing.

Finally, it is also important to point out what ultrasonic measurements do not respond to:

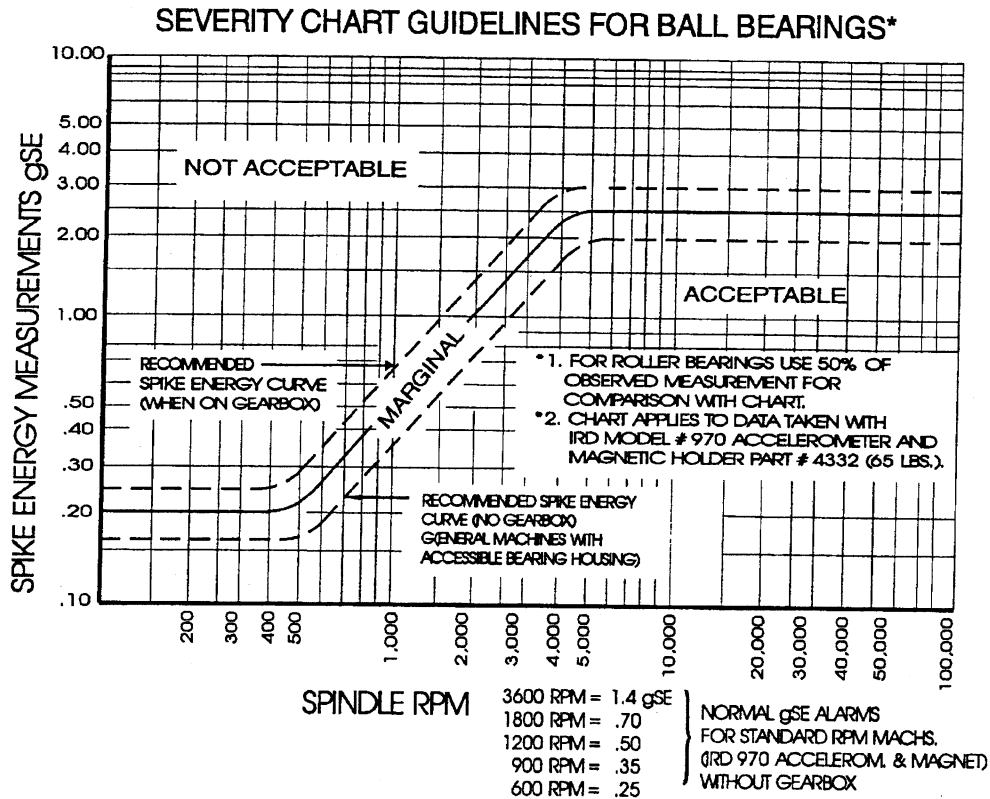
- unbalance
- misalignment
- bent shaft
- electrical problems
- eccentric rotors
- resonance
- structural looseness/weakness
- beat vibration problems

The reason that neither shock pulse nor spike energy respond to the above problems is that they generate lower frequency vibration which is purposely filtered out in the design of their electronics [neither SPM nor overall spike energy are sensitive to problems which generate fault frequencies below about 5000 Hz (300,000 CPM)]. Still, they are very useful in picking up the other problems such as bearing, lubrication, cavitation and rub problems.

Each of these technologies likewise have some disadvantages. In the case of shock pulse, the bearing bore and RPM (or bearing type and number) *must* be known in order to take its reference baseline (otherwise, the results are inaccurate). This is a disadvantage on many machines whose bearing model numbers and bores are not known. Also, since the shock pulse instrument usually makes only ultrasonic measurements, a second instrument is required to obtain vibration readings (particularly *filtered* vibration spectra) which will detect the lower frequency problems. In addition, since the shock pulse instrument is not yet included within any of the major predictive maintenance software offerings, its readings have to be manually entered into these programs if one wants to both store and trend these measurements in his PMP software allowing him to directly compare vibration and shock pulse measurements in one trend graph. (However, SPM does now offer its own software for trending shock pulse readings themselves).

In the case of spike energy, one of its disadvantages currently is that its amplitude response is highly dependent on which specific accelerometer (and mounting) is used to make the measurement. Amplitudes may differ by a factor of 5X to 10X from one accelerometer to another. Thus, if one wants to use current spike energy severity charts like that shown in

FIGURE 3
RECOMMENDED SPIKE ENERGY SEVERITY CHART (IRD SPIKE ENERGY)



SPIKE ENERGY™ MEASUREMENTS

Energy is generated by repetitive transient mechanical impacts. Such impacts typically occur as a result of surface flaws in rolling-element bearings or gear teeth.

This energy is conducted from its source through various paths to the outer surface of the machine structure, and is seen as a small-amplitude vibration at the surface. Accelerometers coupled to the surface generate corresponding electrical signal.

The accelerometer signals processed by unique filtering and detection circuitry to produce a single "figure of merit" related to the intensity of the original impacts. This figure of merit is expressed in "gSE" units.

SPIKE ENERGY™ gSE readings are measurements which can with experience, be correlated with the severity of the casual surface flaws. Even though gSE readings are affected by the nature of the conductive path between the impact source and the accelerometer, similar machine structures will provide a reasonable basis for comparison between the structures.

The gSE figure of merit has proven to be effective in detecting mechanical defects in meshing gears and rolling element bearings. The gSE measurement, when used in conjunction with conventional measurement of vibration velocity and acceleration, provides early indications of mechanical deterioration.

** When used with magnetic holders, accelerometers must be installed with a light coating of silicone grease and tightened to 40 in-lb. torque.

Figure 3, this chart mandates use of the IRD model # 970 accelerometer and model #4332 magnet (65 lbs) to determine whether or not the gSE readings are acceptable. At present, readings from any other accelerometer cannot be used with Figure 3 although work is underway for deriving similar charts for other accelerometer/magnet combinations. One advantage spike energy has over SPM is that one does not have to know the bearing bore and model number to take a reading. However, another disadvantage has been found that when spike energy measurements are taken in the presence of steam, the steam itself can cause variant readings.

The advantages for *both shock pulse and spike energy readings depend on the fact that they mandate that one always takes measurements at identical points from one survey to the next using the same model accelerometer and mounting* (not doing so will cause trends to vary widely). This will improve the accuracy, reliability and repeatability of the data from one survey to the next. In addition, if readings are taken with hand-held probes rather than magnet or stud mounted, simply a change in force on the transducer by the operator can raise or lower the subsequent reading. In both cases, if readings can be taken from stud-mounted locations, trends can greatly be improved. Spike energy measurements have also proven fairly repeatable using magnet mounts (particularly if a high force magnet is mounted on a uniform surface), though not up to the quality and repeatability of those from stud mounts.

FIGURE 4
MAINTENANCE DIAGNOSTIC VIBRATION AND FILTERED HIGH
FREQUENCY GENERAL TOLERANCE CHART FOR PROCESS
MACHINERY WITH ROLLING ELEMENT BEARINGS^①

Unfiltered Vibration Velocity	Shock Pulse & Acoustic Emission	Microlog HFD	IRD Spike Energy ^②	Severity Level and Alarm Status
Inches/Second Peak	dB	g	gSE	
1.5+	50+	5.0+	3.0+	Danger (Shutdown)
.75 - 1.49	40 - 49	3.0 - 4.99	1.50 - 2.99	Very Rough (Alert)
.40 - .74	30 - 39	1.50 - 2.99	.80 - 1.49	Rough (Alert)
.20 - .39	20 - 29	.75 - 1.49	.40 - .79	Fair (Acceptable)
.10 - .19	10 - 19	.30 - .74	.20 - .39	Good (Acceptable)
.01 - .09	1 - 9	.01 - .29	.01 - .19	Smooth (Acceptable)

① Provided by Charles Berggren of Monsanto

② Spike Energy Amplitudes measured using an IRD 970 accelerometer outfitted with IRD's 2-pole, 65 lb. magnet.

In any case, ultrasonic measurements are recommended on high-speed machines or lower speed machines having high frequency vibration sources (rolling element bearings, gears, etc.). They can prove very effective in picking up many problems that might otherwise go unnoticed, particularly during incipient stages, but sometimes even in more advanced stages. In the case of bearings, each of these tools will pick up problems during the first of four definable stages of degradation. Finally, meaningful severity charts are now being developed for each of these high frequency measurement parameters. Figures 3 and 4 are included as example severity charts for spike energy and shock pulse, respectively.

9.2 HIGH-FREQUENCY ACCELERATION (HFD):

Some vendors have offered banded high-frequency acceleration measurements in addition to vibration with their predictive maintenance hardware and software systems. While this is similar to the ultrasonic measurements mentioned above, this is truly a different measurement parameter altogether. In fact, as the name implies, this is a banded high frequency acceleration measurement which measures the total energy existing between certain lower and higher cutoff frequencies. Most often, this high-frequency measurement is referred to as "*HFD*" and is marketed by some of the major predictive maintenance vendors (see "Endnotes" 3 and 4). Figure 4 shows severity levels for "Microlog HFD" as compared with those of spike energy and shock pulse. In general, note that Microlog HFD levels tended to be approximately 2X those of spike energy (that is, about 100% higher in this particular study).

The thing which differentiates HFD measurements from those of shock pulse and spike energy is that HFD has not generally been an ultrasonic measurement, but instead a banded acceleration measurement. In the beginning, HFD by Vendor 3 was normally banded between approximately 5000 Hz and 20,000 Hz (300,000 - 1,200,000 CPM). In recent times, this vendor has allowed the user to specify the HFD lower and higher cutoff frequencies up to as high as 20,000 Hz (1,200,000 CPM). In these cases, it is understood that some users have specified their lower cutoff frequency as low as 1000 Hz up to 5000 Hz and their higher cutoff frequency anywhere from 3000 Hz to as high as 20,000 Hz. "Microlog HFD" (by Vendor 4) provides a fixed frequency range of 5000 Hz to 60,000 Hz (300,000 CPM - 3,600,000 CPM). Like spike energy and shock pulse, HFD is sensitive to faults generating high frequencies such as bearing wear, gear wear, cavitation, etc. Due to its high frequency banding, HFD is not sensitive to such problems as unbalance, misalignment, eccentricity, etc.

Here, it will be of great importance that the analyst consciously uses the same analyzer, transducer and transducer mount (hopefully magnet, quick lock or stud/adhesive mount) in order to make his HFD measurement more repeatable and more reliable.

ENDNOTES:

1. "Spike Energy" - by Entek IRD International Corp. based in Cincinnati, OH.
2. "Shock Pulse" - by SPM Instruments based in Marlborough, CT.
3. "HFD" - by CSI based in Knoxville, TN.
4. "Microlog HFD" - by SKF Condition Monitoring based in San Diego, CA.

9.3 SPIKE ENERGY MEASUREMENTS

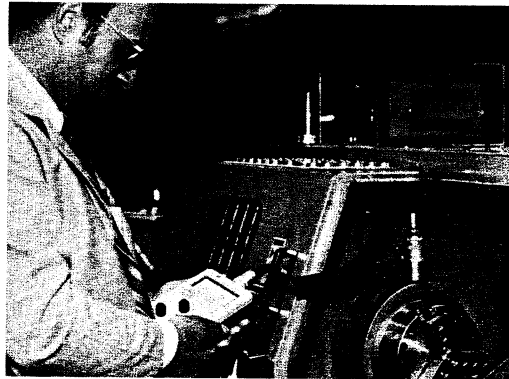
Spike Energy has been defined as "vibration energy generated by short-duration, metal-to-metal impacts and random vibration that propagate through the structure". The following discussion shows how Spike Energy measurements are useful for evaluating the condition of rolling-element bearings.

Bearings are one of the most precisely made devices available, so it follows that the characteristic vibration level of a good bearing is quite low compared to other parts of a machine. During the early stages in the development of a bearing defect, the bearing vibration might still be so low that it is lost in the vibration caused by other parts.

The important point of this discussion is that overall, or filter-out, measurements of vibration velocity or acceleration often do not show the characteristics of a bearing defect until the condition reaches a critical point. So measuring or monitoring the overall vibration might not provide sufficient warning of developing bearing problems.

Figure 6-25 shows an IRD Model 810 Vibration-Spike Energy Detector. This instrument includes provisions for measuring normal vibration displacement, velocity and acceleration. It also includes special provision for measuring Spike Energy levels - vibration that is characteristic of defective bearings and gears.

Figure 6-25
An IRD Model 810 Vibration Meter



As illustrated in Figure 6-26, a defect on the bearing raceways or rolling elements causes intermittent impacts between the bearing components. These brief impacts, in turn, excite the resonant frequencies of the various bearing components. A number of IRD instruments respond to these characteristics of bearing vibration, providing information that is needed for assessing bearing condition.

Figure 6-26
Spike Energy pattern due to a defective bearing

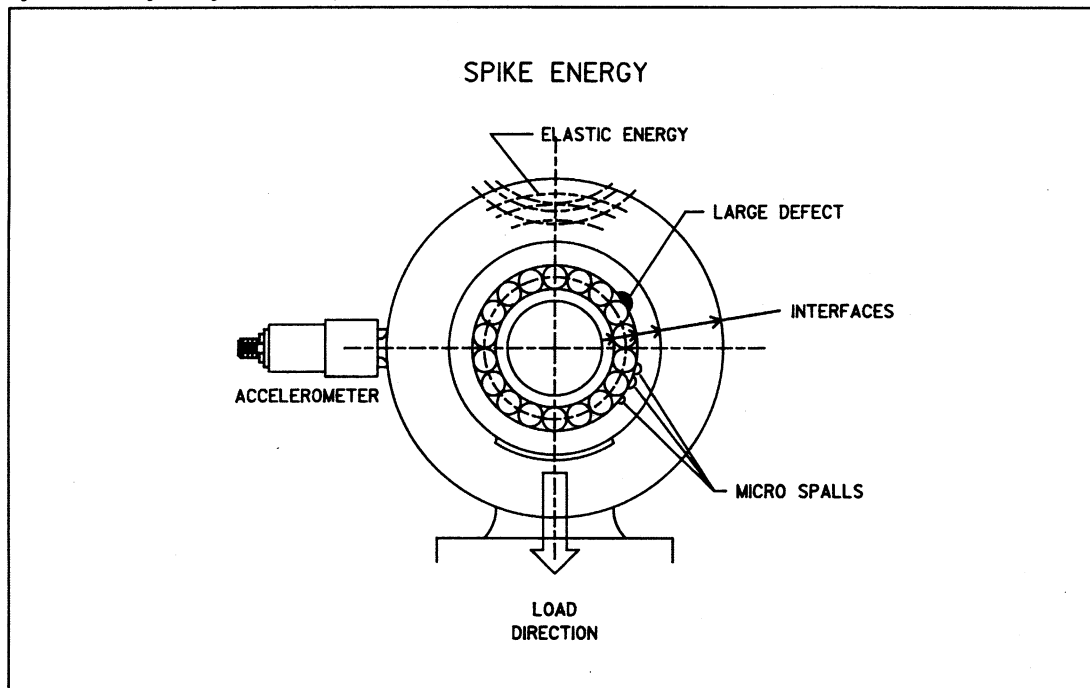


An IRD Model 970 accelerometer pickup has a mounted natural frequency of about 27,000 Hz, or 1,620,000 CPM. This frequency is well above the vibration frequencies generated by mechanical problems in the machine - unbalance, misalignment, electrical problems, aerodynamic forces, hydraulic forces and so on. As a result, the only sources of vibration that excite the natural frequency of this accelerometer are the impact, or spike, forces generated by a defective bearing or gear tooth.

Furthermore, setting the vibration meter to measure Spike Energy introduces a high-pass filter that excludes any remaining frequency components below 5000 Hz (300,000 CPM). Consequently, increases in vibration due to problems such as machinery unbalance and misalignment do not cause any increase in the Spike Energy readings. In other words, the Spike Energy modes of IRD instruments are designed to respond to impact and random vibration forces that are characteristic of defective bearings.

Figure 6-27 shows an accelerometer attached to a machine surface. The energy due to bearing impacts is conducted through various paths to the outer surfaces of the machine. This signal at the accelerometer pickup has a very low amplitude, but special signal-processing circuits in Spike Energy instruments are capable of converting the signal into a numerical quantity, or figure of merit. This value is expressed in g-SE units.

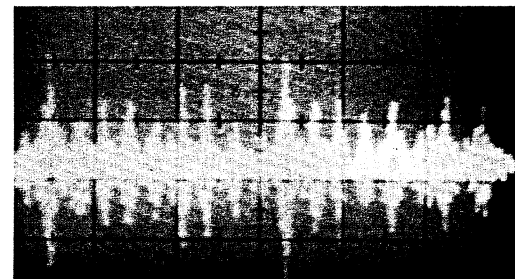
Figure 6-27 Bearing defects generate vibration (SPIKE ENERGY)



Observing the bearing vibration as a time waveform on an oscilloscope can also be helpful in detecting specific bearing frequencies. The time waveform in Figure 6-28 was obtained from a bearing having a known defect on the inner raceway. As the rolling elements pass over the defect, they create the spikes that you can clearly see in the display; and because you note the period of the display, you can readily determine the frequency of vibration. Having determined the frequency of the vibration in this fashion, you can compare it with calculated rotational frequencies of the bearing.

The vibration signal shown on the oscilloscope display varies in amplitude due to the fact that the defect on the inner race is rotating, and thus continually changing position relative to the vibration pickup attached to the bearing.

Figure 6-28
Oscilloscope display of a bearing defect



This variation in the amplitude may appear as regular amplitude modulation of a sinusoidal oscillation or may show somewhat random amplitude variations.

Measurement Procedures

On machines that have no previous history of Spike Energy measurements, you should use the readings for tracking, or trending, changes in bearing condition. A correlation between these Spike Energy readings and bearing failure must be established by actual experience. Once you have established this correlation, however, you have the basis for using Spike Energy measurements to determine the actual bearing condition.

In any case, it is not recommended that you shut down a machine solely on the basis of Spike Energy measurements. You should make this judgment after considering at least one other vibration parameter such as acceleration or velocity. The reason is that periodic Spike Energy measurements can vary by large amounts, because of changes in machine operating condition. Steam flow, turbulence, high pressure air flow, and cavitation, as well as bearing defects, create Spike Energy signals. Therefore, changes in these operating parameters can cause changes in the measured Spike Energy signals.

For example, you might be led to believe that a sudden increase in the Spike Energy reading for a bearing is indicating an imminent failure. But if a subsequent acceleration or velocity measurement shows no significant increase in vibration, you should consider that the sudden rise in Spike Energy levels may be the result of a change in machine operation (e.g., cavitation) rather than bearing deterioration.

The choice of a second criterion for judging machine condition depends on the function of the machine. A very high speed machine is more readily affected by vibrational forces, therefore an acceleration measurement would be a good indicator. For low speed machines, where relative movement affects operation, displacement would be the preferred backup measurement to Spike Energy. Table 6-2 summarizes the relative vibration amplitudes as a function of the measurement parameter and machine condition. Notice that g-SE readings are HIGH for four out of five machine conditions, whereas only two of these conditions actually justify shutdown.

Table 6-2

	VIBRATION LEVELS			
	Disp	Vel	Accel	SE
MACHINE CONDITION				
Machine OK	OK	OK	OK	OK
Watch Machine Bearing Defects OK to Run	OK	OK	OK	HIGH
Watch Machine Bearing Defects OK to Run	OK	OK	HIGH	HIGH
Machine Problems Analyze/Shutdown	OK	HIGH	HIGH	HIGH
Machine Problems Analyze/Shutdown	HIGH	HIGH	HIGH	HIGH

9.4 HIGH FREQUENCY ENVELOPING AND DEMODULATION TECHNIQUES

Various vendors provide different high frequency spectrum enveloping techniques. Not all of them will be covered in this manual at this time. The following is a list of their titles:

- a. gSE Spectrum (By IRD Mechanalysis based in Columbus, OH)
- b. Acceleration Enveloping (By SKF Condition Monitoring based in San Diego, CA)
- c. SEE (Spectral Emitted Envelope) (By SKF Condition Monitoring based in San Diego, CA)
- d. Amplitude Demodulated Spectrum (By CSI based in Knoxville, TN)

Following in Section 9.41 will be a discussion of one of these offerings (gSE Spectrum). Section 9.5 includes example of data taken using both gSE Spectra as well as other of the above parameters.

IRD FAST TRACK.

gSE SPECTRUM.

BY STUART COURTNEY.
MANAGER TECHNICAL SERVICES.
IRD MECHANALYSIS (UK)LTD.

INTRODUCTION TO gSE SPECTRUM

Industrial equipment downtime is often caused by bearing and gear failures. Therefore, it is important that these types of failures be considered in any machine condition monitoring or predictive maintenance programme. The signals caused by bearing and gear faults are often low in amplitude and jumbled together, this makes them extremely difficult to separate out of a spectrum for detailed analysis. Spike Energy™ Spectrum (gSE Spectrum), an already proven technique incorporated in IRD portable instruments. It is used to assess rolling element bearing and gear condition and is the key to analysing these low amplitude signals.

The overall gSE amplitude is obtained by a unique filtering process. It measures the INTENSITY of energy of the minute impacts that occur when an antifriction bearing rotates. These impacts are measured in the ultrasonic frequency range i.e. above 5 KHz (300,000 CPM), the resultant NUMBER is a Figure of Merit that can be used as an assessment of a bearing's condition.

Spike Energy Spectrum is used to provide more detailed information about bearing and gear defects than the overall gSE indication.

In the normal narrowband spectrum there are many components that can be considered as quite normal and acceptable. The primary purpose of gSE Spectrum is to detect which element(s) of a bearing or gearbox may be deteriorating, the vibration energy of these faults may be hidden in the normal satisfactory spectrum.

Spike Energy Spectrum is a reliable measurement used for machinery diagnostics. It is also effective on low speed machinery such as paper rolls and on machines where there is a high degree of background noise such as may be experienced in reciprocating machines or cavitating pumps.

DISCUSSION

The method of signal processing used for gSE Spectrum provides an enveloping and demodulation technique which can identify periodic impacts. These key signs of a deteriorating rolling element bearing are independent of typical vibration signals. This works in exactly the same way that an audio signal is extracted from the carrier frequency in AM radio. Another advantage of the gSE Spectrum technique is that it can be used to distinguish between the different faults associated with the individual components.

Before we can discuss the method of analysis used for gSE Spectrum it is important to remember that the forces produced by the normal low frequency defects are generally in the form of a steady push and are sinusoidal in their nature.

When a defect develops on an internal surface of a rolling element bearing, bursts of energy are generated each time contact with the defect is made. This information will be present in the measured vibration signals and the bursts will be repetitive at a rate determined by the type of defect, the bearing geometry and the speed of the machine. The frequencies of these bursts are very often the same (or nearly the same) frequencies as that of the steady push frequencies generated by the unbalance, misalignment etc.

These repetition rates are known as the bearing defect frequencies. There are 4 main defects associated with the rolling element bearing :

- (1) Ball passing frequency - outer race (BPFO)
- (2) Ball passing frequency - inner race (BPFI)
- (3) Ball spin frequency (BSF) for a fault on the ball itself.
- (4) Fundamental train frequency (FTF) for a cage fault

These characteristic frequencies in CPM can be calculated as follows:

$$\begin{aligned} \text{BPFO} &= 1/2 \cdot \text{RPM} \cdot N \cdot (1 - \text{Bd}/\text{Pd} \cdot \cos \phi) \\ \text{BPFI} &= 1/2 \cdot \text{RPM} \cdot N \cdot (1 + \text{Bd}/\text{Pd} \cdot \cos \phi) \\ \text{BSF} &= 1/2 \cdot \text{RPM} \cdot \text{Pd}/\text{Bd} \cdot (1 - (\text{Bd}/\text{Pd} \cdot \cos \phi)^2) \\ \text{FTF} &= 1/2 \cdot \text{RPM} \cdot (1 - \text{Bd}/\text{Pd} \cdot \cos \phi) \end{aligned}$$

Where RPM = rotating frequency, CPM
 N = number of rolling elements per race
 Bd = rolling element or ball diameter, inches (mm)
 Pd = rolling pitch diameter, inches (mm)
 ϕ = contact angle, degrees

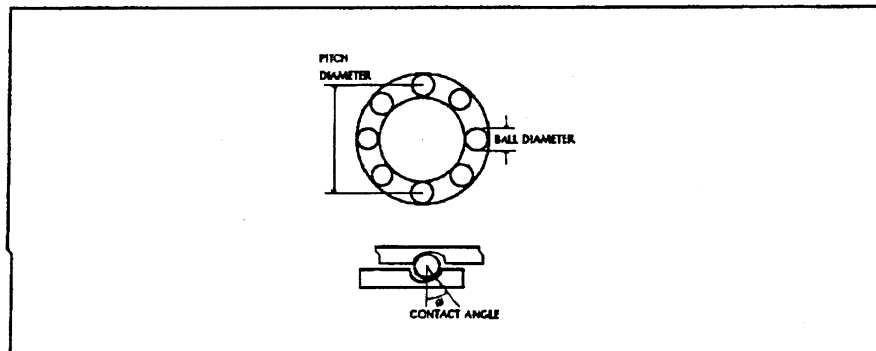


Figure 2 ROLLING BEARING GEOMETRY.

All of these detected frequencies can be similar to many other frequencies found in the rotating elements of a machine. As these other defects can generate much higher signal levels, it becomes a very difficult task to extract meaningful bearing information. Bursts generated by bearing defects tend to excite natural frequencies of the machine structure (i.e. bearing block) or the transducer. Hence, a signal resulting from these bursts (or spikes) appears as periodic bursts of high frequency energy at intervals determined by the actual bearing defect. In effect, the bearing defect frequency modulates the resonant frequency of the system and the result is a very low level high frequency component. Thus, the spectral waveform may show only a low level amplitude at the resonance frequency and it is typically lost in the background noise.

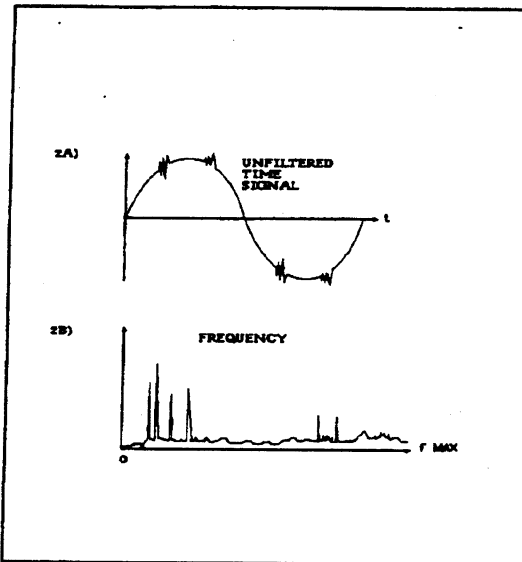


Figure 2 UNFILTERED TIME SIGNAL.

Getting to the details of the defect frequencies requires a three-step process. First, the time domain signal is filtered via a band-pass filter with a selectable low frequency corner. It can be seen in 3A that the low frequency sinusoidal type components are removed from the Time Waveform and if a spectrum analysis was carried out on this information it would simply remove the low frequency components as shown in 3B.

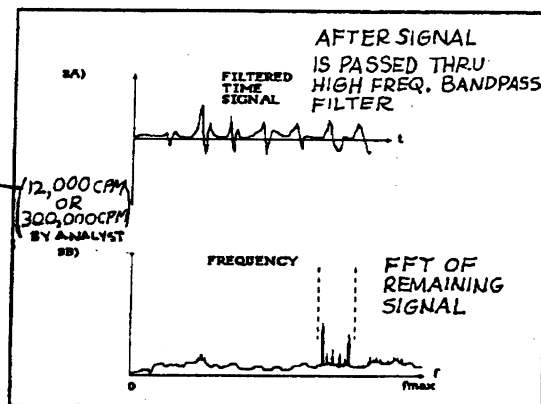


Figure 3 FILTERED TIME SIGNAL.

In order to identify the low frequency repetition type signals that cause the high frequency excitation of the machine components, the signal is routed through a peak-to-peak detector to extract the repetition rate relating to the particular bearing defect.

Lastly, the spectrum of this demodulated signal is developed to show the particular bearing defect frequency and its harmonics.

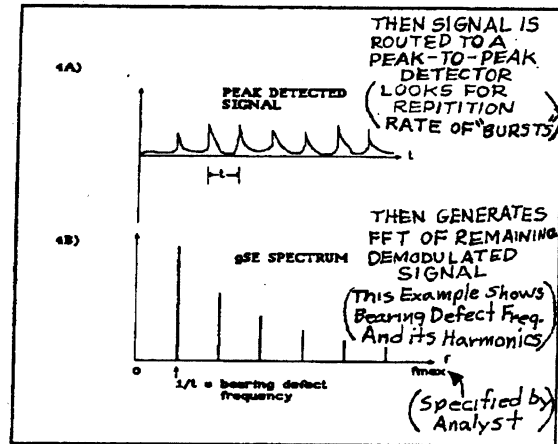


Figure 4. PEAK DETECTED SIGNAL AND SPECTRUM OF gSE

The Spectrum in Figure 4a. is a gSE Spectrum taken with the IRD FAST TRACK DATA COLLECTOR; it can be seen that the impact energy caused by a the Ball Pass Frequency Inner of an SKF type 206 bearing in very early stages of failure. Harmonics are present and there is no sign of the low frequency steady push vibration that would be present in the narrow band spectrum. It is interesting to note that the frequency corresponds to the 8 x fundamental caused by the vane pass frequency, this would mask the bearing frequencies in the narrow band spectrum. This method of signal processing obviously does not replace the narrow band spectrum method of analysis. gSE Spectrum is an extra tool in the hands of the vibration analyst that allows assessment of the condition of a machines bearings and gear components.

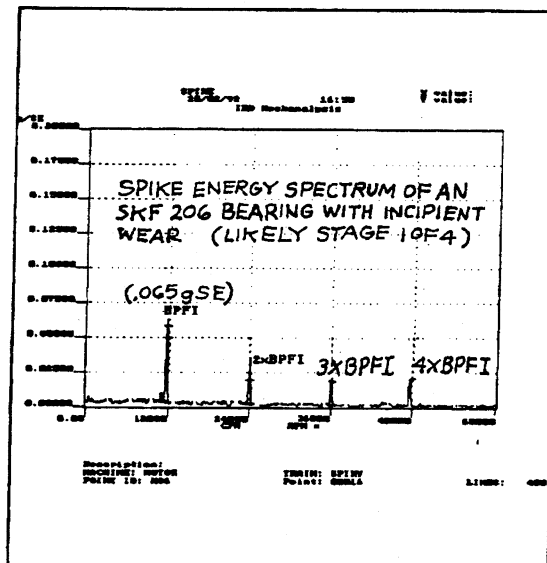


Figure 4a gSE spectrum of an SKF 206.

FAST TRACK SETUP PARAMETERS.

The uniqueness of the IRD gSE spectrum method of signal processing allows the operator to choose the setup menu that most suits the application.

The gSE spectrum setup menu allows selection of the High pass corner frequency.

The **LOW** corner frequency sets the filter to 200Hz. This setting will allow detection of defects on low speed bearings such as paper rolls. (12,000 cpm)

When the **STANDARD** corner frequency is selected, the filter is set to 5KHz. (300,000 cpm)

The **DECAY TIME CONSTANT** is used to ensure that accuracy of measurement is maintained for all measurements. It is automatically switched depending on the chosen FMAX.

SUMMARY.

In general, the first detectable signs of bearing deterioration are evident in a gradual increase in vibration levels above 2KHz. The faults make contact with other surfaces and structural or transducer resonances are excited and appear as part of the spectrum. These bursts (spikes) cause a modulation frequency that is directly related to the fault causing it. Once the analog signal is passed through a band-pass filter with the appropriate low frequency corner and demodulated by the peak-to-peak detector, the resulting spectral information can be analyzed in detail to determine the specific bearing component deterioration.

As a "normal" rolling bearing will exhibit some vibration, care must be exercised to accurately determine and assess the severity of bearing deterioration. This can be done with personal experience, machine history and by establishing a threshold level for severity assessment. Once a noise background level has been established, the spectral amplitudes can be examined to determine the onset of bearing deterioration and to establish the required corrective action.

Using an IRD **FAST TRACK DATA COLLECTOR** it is a relatively easy task to make these gSE Spectrum measurements to determine early signs of bearing and gear failures. However, it is important to adhere to a few rules. First, all measurements should be made with an accelerometer that is stud mounted. If this is not possible, a 35 Kg magnet mount is acceptable. Second, the IRD Fast Track Data Collector/Analyzer uses selectable high pass filter settings. The selected high pass filter frequency depends on the overall noisiness of the vibration signal. As a rule of thumb, the lowest cutoff frequency that will extract the signal envelope is the best, as it will not be as demanding on the selected transducers performance. (65/b)

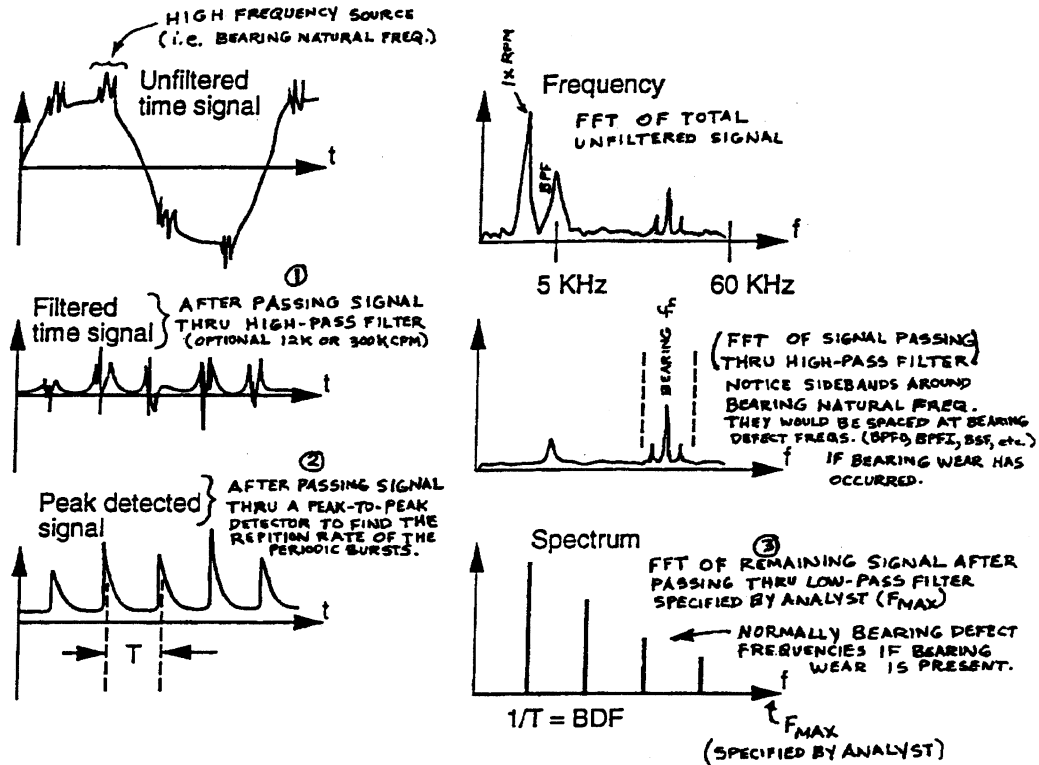


ILLUSTRATION B
SPIKE ENERGY SPECTRUM 3-STEP PROCESS
(PERFORMED ON THE TIME WAVEFORM)

9.5 CASE STUDIES

9.51 TESTIMONIAL ON THE VALUE OF SPIKE ENERGY SPECTRA

August 21, 1992

Technical Associates of Charlotte, Inc.
347 North Caswell Road
Charlotte, NC 28204

Attention: Jim Berry

Dear Jim,

Enclosed are the velocity and spike energy spectra you had asked for. There are a total of four different machines with a velocity and spike energy spectrum for each one. Each machine had a damaged bearing as indicated by the spectra, but it should be noted that the defects were no more than a stage two bearing failure. I personally feel that if the bearing were allowed to run longer, the velocity spectrum would have indicated the bearing flaw to a greater degree. I am pleased, however, with the spike energy spectrum as an early bearing flaw indicator. I have received and installed the latest Entek software update version 3.27, which supports the spike energy spectrum in the route mode, and hope to soon have a database of gSE spectra to help improve the predictive maintenance program at our mill.

If I can be of further assistance to you, please let me know.

Sincerely,

PAPER MACHINE

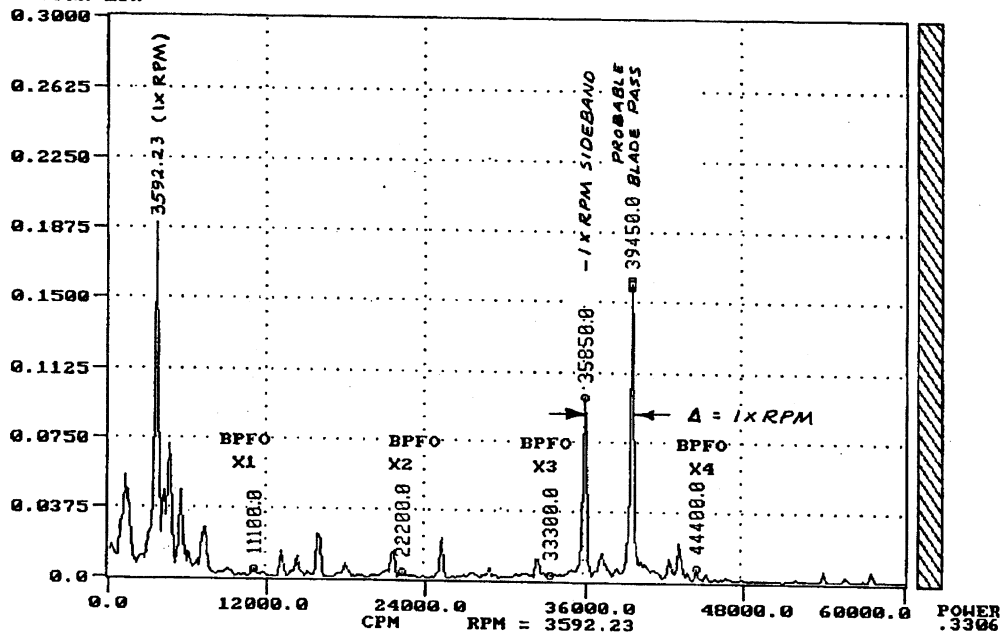
06/26/92

9:17:00 AM

X value: 39450.00

Y value: 0.159797

N/S Peak Lin



Description: 811.2117

MACHINE: ISO VAC BLOWER

TRAIN: MEZZANINE

2 Vertical

POINT ID: MTR DE U

Point: 01853

PAPER MACHINE

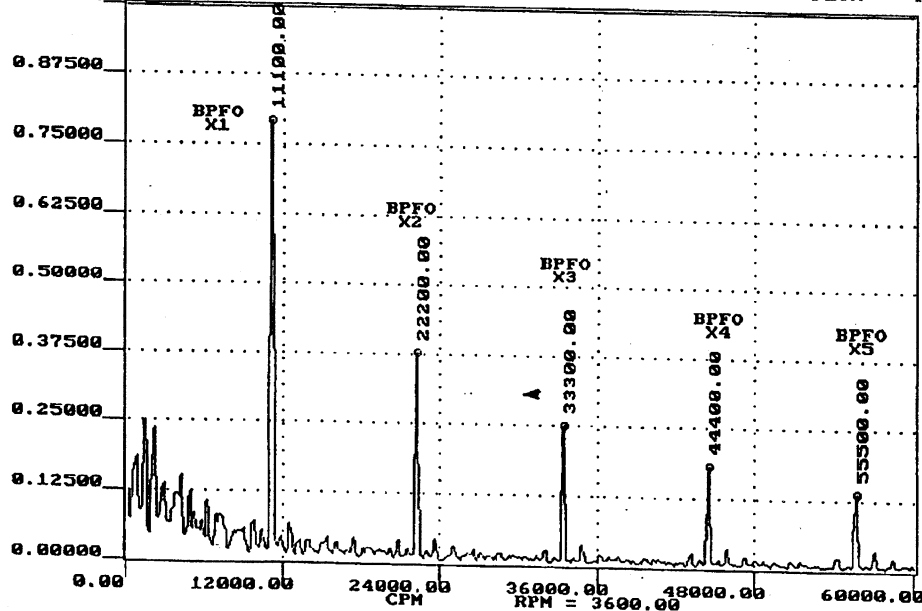
06/26/92

01:26:00 PM

X value: 55500.000

Y value: 0.1372099

AMP RPM USER AMP USER HDBANKTAKING
g/SE Hz I DISP L DISP COUNT FPM
PHASE PHASE DEG-F DEG-C CNT RT MPM Vdc Vdc
PRESS PRESS FLOW FLOW



Description: 811.2117

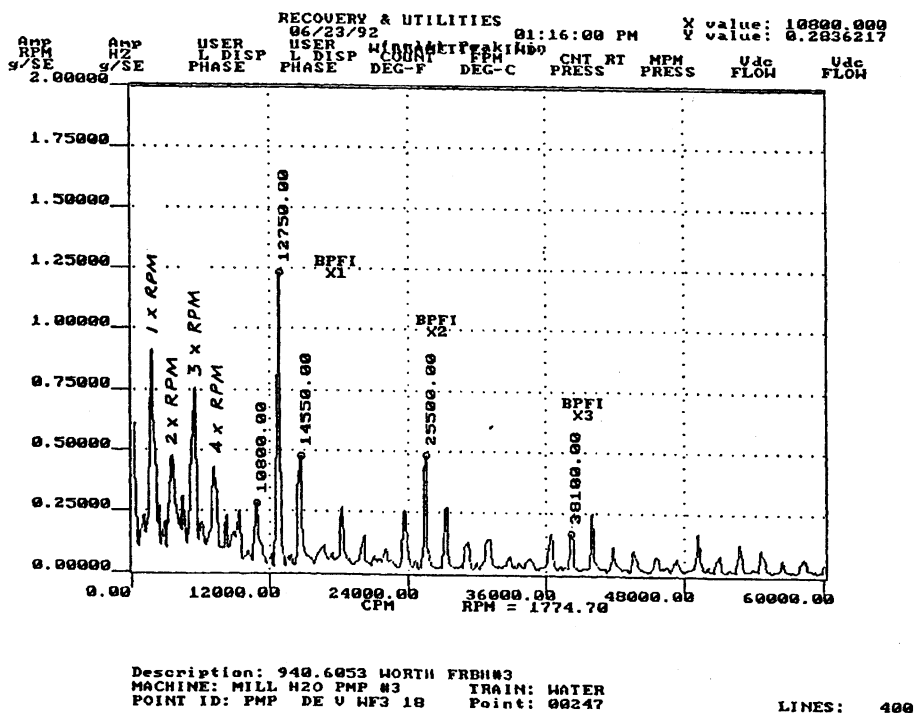
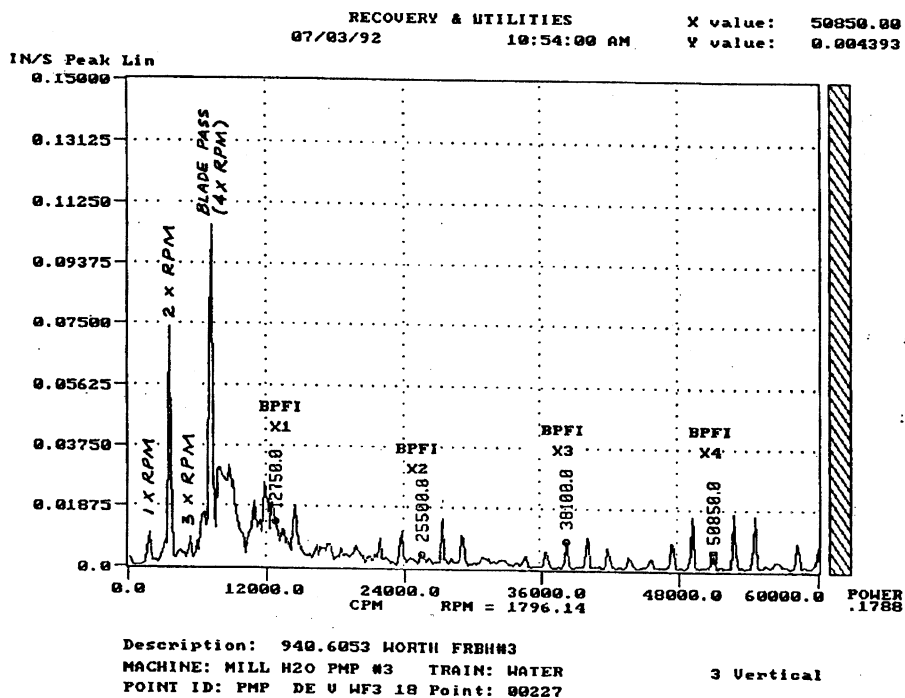
MACHINE: ISO VAC BLOWER

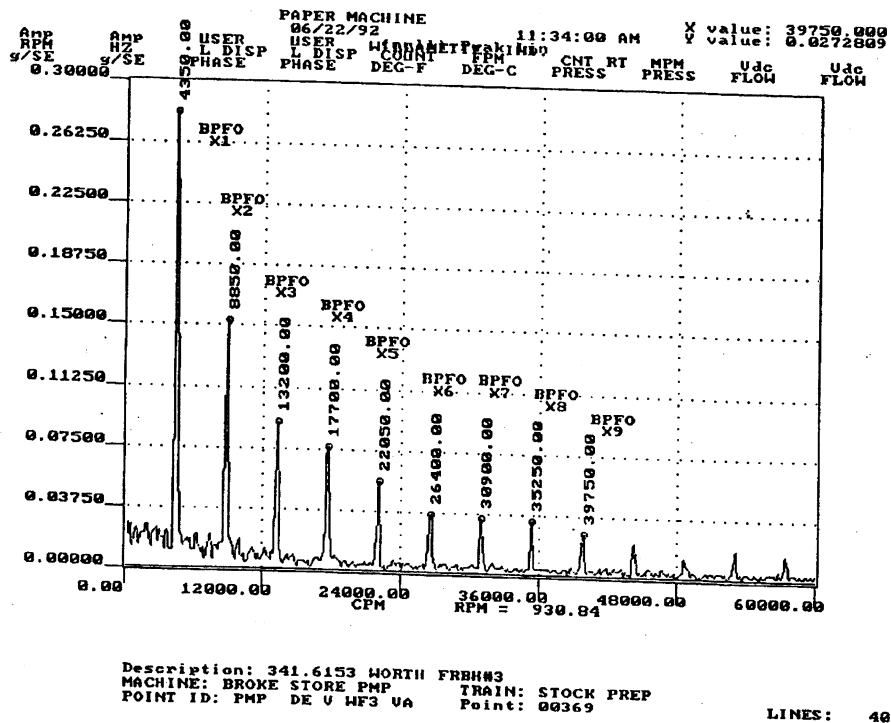
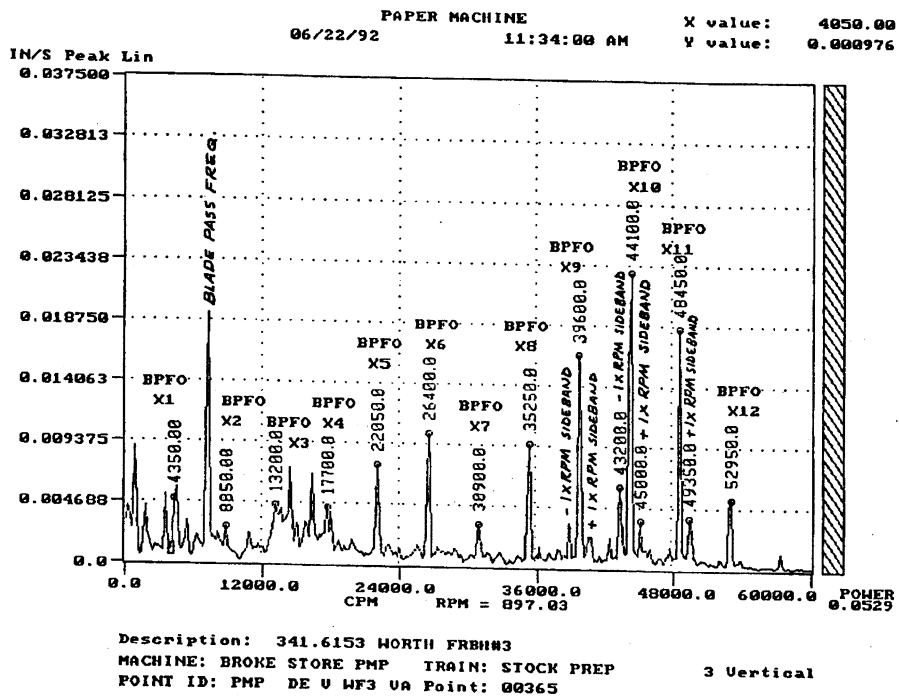
TRAIN: MEZZANINE

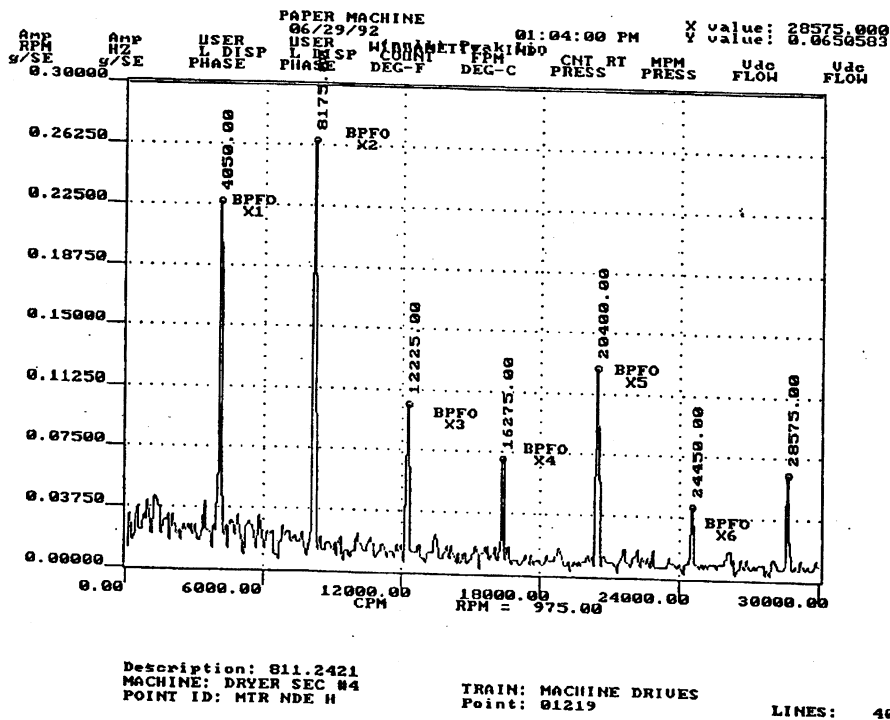
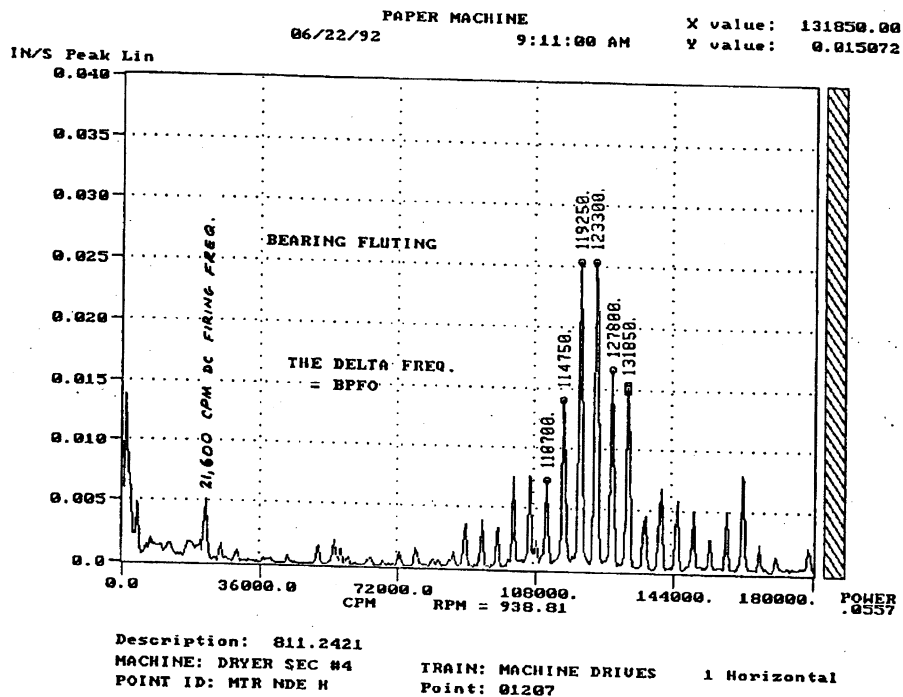
LINES: 400

POINT ID: MTR DE U

Point: 01854



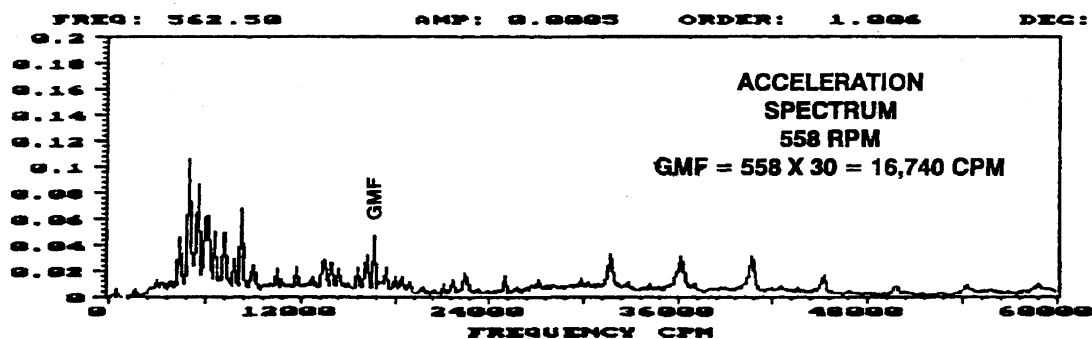




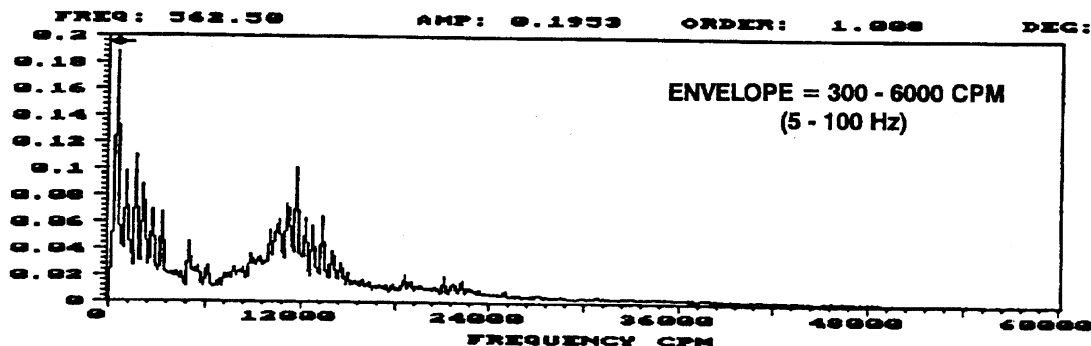
9.52 ENVELOPING (CASE STUDY #2):

Acceleration Enveloping is an extremely useful technique for detecting impulse producing faults such as bearing and gear mesh defects. It is very important to select the proper bandpass filter to facilitate proper implementation of the technique. The suggested lower cutoff of the bandpass filter is at least 10X running speed. This case shows the importance of the proper filter selection for Acceleration Enveloping.

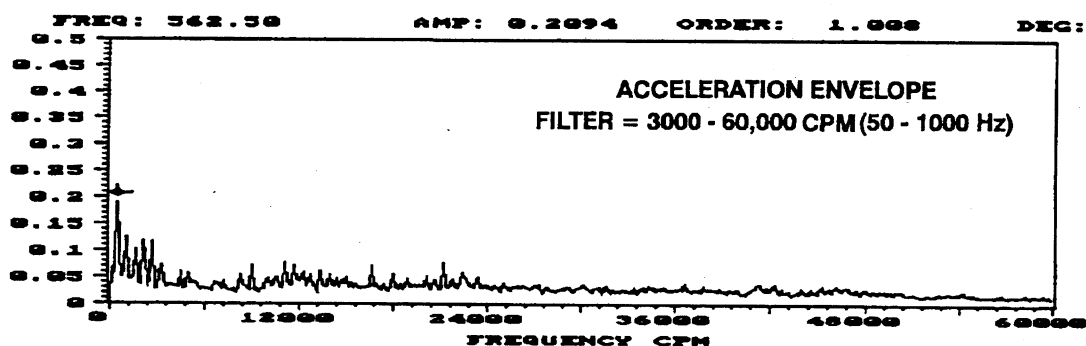
Case 2 monitors a Paper Machine Dryer Pinion with 30 teeth rotating at 558 RPM. As can be seen from the 6400 line acceleration spectrum below, a clear fault cannot be observed in the vibration data. The main observations which can be made are (1) harmonics spaced at running speed from about 2500 CPM to about 21,600 CPM; (2) other blunt peaks in the upper range of the spectrum.



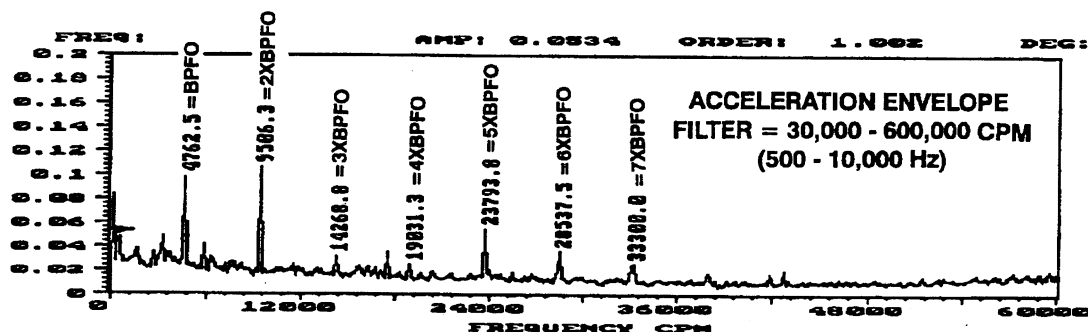
Acceleration Enveloping was developed to separate machinery faults which generate vibration impulses such as bearing faults and gear meshing problems, from large carrier signals which are often associated with normal machinery operation. Below is a spectrum showing an Enveloped Acceleration Signal using the 300 CPM to 6000 CPM envelope. Normally the envelope is chosen such that the low end cut off is at least 10X greater than the running speed. At a running speed of 558 CPM, this would mean the envelope should have a lower range of at least 5580 CPM. In this case, little enhancement of the signal can be seen in the vibration signal if a fault were present.



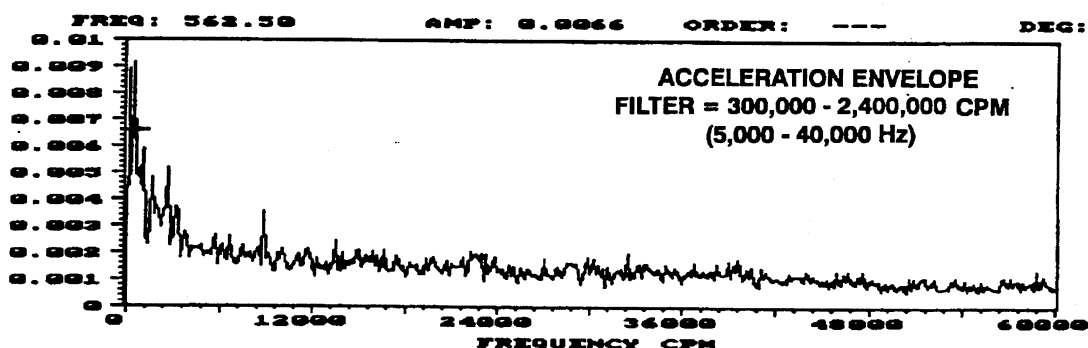
The following figure shows the enveloped signal using the 3000 CPM to 60,000 CPM bandpass filter range. The lower end of the bandpass filter is about 5X running speed. Again this fails the criteria for the lower range of the bandpass filter to be above 10X rotational speed.



Clearly, the following enveloping filter range, 30,000 CPM to 600,000 CPM, is the proper range for this signal, as only peaks relating to an outer race defect are evident. This is also the first filter range with a lower cutoff value greater than 10X the nominal running speed. *The signal is much clearer than previous and analysis shows harmonics of 4762.5 CPM relating to an outer race defect for an SKF 22228CCK bearing.* The low amplitude enveloping signals at the calculated outer race defect frequency indicate early detection of an outer race defect.



The final spectrum is an enveloped signal using the 300,000 to 2,400,000 CPM bandpass filter. The lower range of the bandpass filter is over 500X running speed and clearly unacceptable for an enveloping bandpass filter range.



The bearing was removed from service during a down week for the paper machine and inspection showed some water damage and surface cracking in the outer race. During normal operation this bearing would have been left in service until trend levels had increased, but as scheduled down time is opportune, it was removed.

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